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The VASE pedagogical framework

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THE VASE PEDAGOGICAL FRAMEWORK: TEACHING FOR VALUES IN DESIGN IN HIGHER EDUCATION

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ABSTRACT

This chapter introduces and describes a pedagogical framework and resources supporting teachers in teaching for values in design. The pedagogical framework along with its teaching activities and resources was developed through the project Value Sensitive Design in Higher Education (VASE) to facilitate the development of knowledge, skills, and attitudes in students across the Humanities and Sciences that help them become more responsible designers. The VASE pedagogical framework is built around three core competency pillars reflecting the identified central dimensions involved in teaching for values in design: 1) Ethics and Values, 2) Designers and Stakeholders, 3) Technology and Design, and provides teachers with a curriculum compass, a collection of 28 teaching activities and 12 assessment activities. The chapter presents this work as an inspirational repository for all teachers to explore, experiment with and integrate in their teaching for values in design.

PEDAGOGICAL FRAMEWORK, TEACHING & ASSESSMENT ACTIVITIES, STUDENTS AS RESPONSIBLE DESIGNERS, PEDAGOGICAL DESIGN PATTERNS, OPEN EDUCATIONAL RESOURCES

1. INTRODUCTION

Teaching for values in design is currently gaining momentum as addressing ethical and social dilemmas is increasingly becoming a part of the learning goals in a growing number of university courses around the world (Hendry et al., 2020). This increased interest stems in part from ethical issues and scandals surrounding tech industries becoming more pervasive in the media and public discussions (Fiesler, 2020; Singer, 2018). However, as researchers, developers, and teachers within the field of technology and design, we find that educational resources and materials to support such teaching are still scarce, and that many teachers like ourselves face the pedagogical challenge of how to teach and make explicit the values perspective in the classroom.

To provide teachers with resources enabling them to educate students on these matters, this chapter presents a proposal for a pedagogical framework aiming at facilitating the development of knowledge, skills and attitudes that students need in order to consider the broader context and implications of design and design processes, and through this the possibility for them to become more responsible designers. The pedagogical framework proposed in this chapter can be used for:

- scaffolding students' value-sensitive knowledge, skills and attitudes and capabilities to work with values in design
- assisting teachers in how to teach students to become more responsible designers (a so-called double-pedagogical framework, teaching teachers how to teach for values)
- capacity building for higher education institutions by supporting institutions in educating responsible designers

The pedagogical framework was developed through the project Value sensitive design in higher education, VASE (2021a), which involves a collaboration between teachers, researchers, and educational developers at Malmö University (Sweden), Aarhus University (Denmark), and Eindhoven University of Technology (the Netherlands). The pedagogical framework (from now on referred to as the VASE framework) and the resources developed are offered as an open educational resource (OER), the VASE OER online platform (VASE, 2021), to teachers at different technology, design, and engineering programmes such as interaction design, information science, computer science, technology enhanced learning, educational technology, industrial and product design, or similar programmes, courses, and modules. The reason for targeting such a broad array of programmes and disciplines is because of the multidisciplinary nature of designing for and with values. Educating students to become responsible designers and supporting teachers in how to teach for this, is equally important for all engaging

with technology and design across the Humanities and Sciences. As such, the VASE framework targets the 'meso level' of teaching and learning resources, implying that it is neither a 'macro frame' for what values in design are and how they should be understood as a discipline, nor a 'micro activity' for how to teach within a specific curriculum. All in all, the VASE framework reflects the multidisciplinary fields of the researchers, developers, and teachers who came together to develop the framework and its pedagogical resources. As such, the hope is the framework will help people like ourselves integrate teaching for values in design in meaningful ways within their own diverse disciplines, programmes, and modules.

The VASE framework is built up around three core competency pillars reflecting the identified central dimensions involved in teaching for values in design: 1) Ethics and Values, 2) Designers and Stakeholders, 3) Technology and Design. It has seven overarching learning objectives. To help teachers navigate and plan a desirable learning pathway that fits their own education programme, we developed a curriculum compass, outlining core competencies to help students work towards becoming more responsible designers. The curriculum compass is on the 'meso level' and, thus, general enough to be applied across various forms of teaching design in higher education. Yet it provides a concrete and practical tool for teachers to navigate through a wide range of useful resources to identify specific teaching and assessment activities that fit their educational contexts and pedagogical aims. The goal is not to provide a full curriculum or courses on values in design, but rather an inspirational repository of various resources for teachers to explore, experiment with and integrate in their teaching based on their particular needs and settings.

The design of the VASE framework is rooted in previous research addressing the importance of creating an increased awareness of the role that values play in design, including initiatives such as value sensitive design (Friedman & Hendry, 2019), values in design (Nissenbaum, 2005), values at play (Belman et al., 2009), and values-led participatory design (Iversen & Leong, 2012). These approaches have been chiefly developed for research and development purposes, offering methods for designers to work intentionally and practically with values in their design practices. Here, the VASE framework adds yet another layer, namely the pedagogical perspective, and how we as teachers can create conditions for students to grow into responsible designers of future technologies.

2. METHOD: DEVELOPING THE VASE FRAMEWORK

We are witnessing an increased interest in teaching and learning about ethics and values in design (Fiesler, 2020; Singer, 2018). However, as a group of experienced teachers in design in higher education, we felt that the teaching resources on this topic are scarce and scattered, which is why we decided not only to develop practical teaching materials, but also to package them into a coherent pedagogical framework.

The development of the VASE framework is grounded on research and methods that combine desk research, teaching practice, educational design processes, pedagogical design pattern development, and peer-review shepherding to secure research-grounded, method-driven, and quality-assured activities for teaching for values in design. One of the core methods that we have applied in this development is the Structure of the Observed Learning Outcome (SOLO) taxonomy (Biggs, 1982). SOLO taxonomy is a five-tier hierarchical framework for structuring learning outcomes, which provides a general tool for defining curriculum objectives, intended learning outcomes, and evaluating learning outcomes. Each level in the SOLO taxonomy is represented by a set of verbs that can be used to formulate intended learning outcomes for each of the teaching activities, such as 'identify', 'describe', 'relate', and 'reflect'. These SOLO learning outcomes clarify the level of competency that is intended to be achieved through the teaching activity. To give a few examples, in the VASE framework, teachers will find learning objectives such as 'Students will be able to analyse, compare, and argue for how values are manifested in design', 'Students will be able to interpret, adapt and plan their design direction based on the identified designer and stakeholder values', and 'Students will be able to critically reflect on the evaluation of the design in terms of values'.

Overall, the development of the VASE framework was guided by the three-phased model for conducting educational design research and developing educational materials (McKenney & Reeves, 2012). Educational design research is a type of educational research aimed at providing concrete solutions to educational practitioners in relation to practical and complex educational problems such as how teachers can teach for values in design and educate students as responsible designers. Solutions can take the form of educational products, processes, or programs that both support teachers in their educational practice and seek to discover new knowledge that can inform future research, development, and practice within that domain. Below is a short description of the three-phased model:

PHASE 1) EXPLORATION AND ANALYSIS

Exploring the existing domain of teaching for values in design by conducting a literature review on (teaching and assessing) values in design and through this developing the research grounding of the VASE project presented in several desk research reports. Furthermore, the method of pedagogical design patterns as a systematic educational development method was introduced. This method was used to develop the concrete teaching and assessment activities such as Values clustering for developing value vocabularies, Constructing value-based design requirements, or Public examination of values on design that can be found on the VASE OER online platform (VASE, 2021).

PHASE 2) DESIGN AND CONSTRUCTION

Based on phase 1, three core competency pillars and seven overarching learning objectives were extrapolated and described. Then the SOLO taxonomy (Biggs, 1982) was applied to the pillars and learning objectives to describe how we as teachers can develop students' competencies from a beginner to an advanced level. This led to the construction of the curriculum compass (Figure 1) as an overarching model for how teachers can plan, carry out and evaluate teaching and learning for values in design. In parallel, several rounds of so-called pattern mining workshops were conducted on the desk research reports and amongst the project participants to capture existing knowledge and practice. Through this process, a large catalogue of potential teaching activities was generated, and from this catalogue, a selection of potential activities (pedagogical patterns) was selected and developed resulting in the design of 28 concrete 'meso level' teaching activities cutting across disciplines and spanning the three pillars and different SOLO levels. Each teaching activity can be selected and used as a standalone activity or in combination with other activities. Furthermore, 12 assessment activities were developed using the same method, aimed at supporting teachers in evaluating whether the intended learning outcomes were achieved by the students.

PHASE 3) EVALUATION AND REFLECTION

Alongside the design of the 28 teaching activities and the 12 assessment activities, iterative peer-review of all activities were carried out. For this, a pedagogical pattern evaluation method called shepherding (a method very similar to peer-review processes within research publications, see Harrison, 1999) was used. This method ensured multiple cycles of evaluation, reflection, and revision of the activities throughout the project. Parallel to this, the teaching activities were put into practice and evaluated through 38 pilots involving around 50 teachers and 1,563 students coming from

various disciplines, programmes, institutions, and countries. The VASE project's development process concluded with the publication of all teaching and assessment activities, the curriculum compass, cases, reports, and research publications on the VASE OER online platform (VASE, 2021), which combined offers an expansive pedagogical framework for teachers teaching design and technology subjects in higher education.

As described above, the main outcome of the project – the collection of teaching and assessment activities – was developed using a modification of the pedagogical design pattern method (Goodyear, 2005; Nørgård et al., 2019, 2022; Köppe et al., 2018; Laurillard, 2012). The method is aimed at capturing 'best practice' from research and practice, which are then developed into concrete 'meso level' activities for teaching and learning within a specific domain; here teaching for values in design. The pedagogical pattern method has been modified in that we have iteratively developed our own pattern template (Table 3). Some of the key differences include: explication of students' learning objectives based on the SOLO taxonomy to enable teachers to target different learning levels when teaching for values in design; development of both teaching activities and assessment activities for teachers to provide them with methods for assessing learning in relation to values in design; a strong focus on describing the concrete steps teachers and students take during the activities to support experienced as well as less experienced teachers within the practice of teaching for values in design.

As teachers ourselves, on the one hand, we find that teaching activities need to be sufficiently elaborated so that we know what to do, how to do it, and why we should do it. On the other hand, the activities must not be seen as 'micromanaging' but instead be open enough to adapt to the way we teach for values in design within our own specific disciplines, programmes, or modules. This is important to ensure the adaptability and reusability of the offered teaching activities across different contexts and disciplines while preserving the best practice of the activity. Here, pedagogical design patterns offer a systematic way for developing new teaching and learning activities through creating couplings between educational theories, methods, and practice (Goodyear, 2005). The core premises of developing a substantial and sound pedagogical pattern collection (here: teaching and assessment activities) are 'systematisation', 'sharing' and 'adaptability' (Goodyear, 2005; Laurillard, 2012; Pedagogical Patterns Project, 2012; EuroPLoP, 2021). Systematisation of educational development and practice involves reflective and iterative development processes where all activities are designed using concerted methods, identical templates, and joint review procedures. Sharing of the systematically developed collection happens through creating an openly published activity repository or teaching resource (here: the VASE OER platform published under Creative Commons) to be shared with all teachers and educational developers within that domain. Adaptability is achieved through ensuring that the descriptions and instructions of each activities targets the 'meso-level', i.e., the activity contains enough information for non-experts to be able to carry out the activ-

ity but without the activity taking over or 'micromanaging' the teaching situation (here: the VASE activities can be taught across a range different disciplines and educational contexts and based on different pedagogical purposes and approaches).

3. RESULTS: THE VASE FRAMEWORK

The visualisation model of the VASE framework (Figure 1) maps the teachers' movement between the different dimensions of the framework when they are planning and carrying out teaching (moving from the centre of the model outwards) or are checking for students' learning (moving from the edge of the model inwards). The hope is that the VASE framework, on the one hand, can support teachers to systematically plan and carry out teaching, and, on the other hand, can scaffold robust learning by giving teachers ways to assess and check learning outcomes (these step-by-

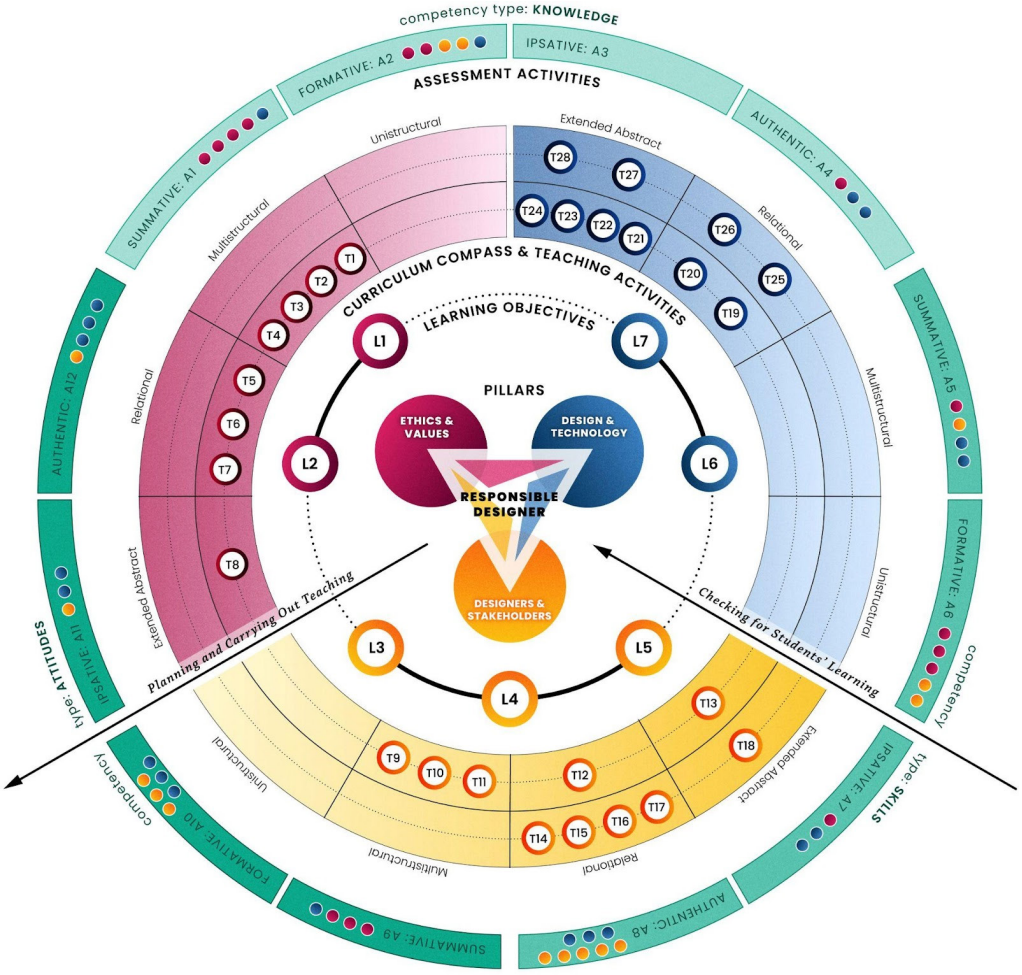


Figure 1. The visualised VASE pedagogical framework for planning and carrying out teaching (moving from the centre of the model and outwards) or checking for students' learning (moving from the edge of the model and inwards).

step processes are put into more detail below). Furthermore, the framework aims at advancing such processes through a multiuse approach to ensure both pedagogical flexibility and contextual adaptability. For teachers, the VASE framework can be used to select relevant stand-alone activities targeting specific values in design dimensions or learning objectives that are the most relevant for teachers within a particular discipline, curriculum, module or teaching session. But it can also be used to create longer in-depth learning pathways for students through combining several teaching activities within a specific domain, that develop students' understanding from simple (unistructural) to complex (extended abstract). Furthermore, teachers can also combine activities across different design dimensions and learning objectives to provide students with a broad foundation to values in design within an appropriate level of competence.

The VASE framework is constructed from and organised around different elements that, when taken together, constitute the main results of the VASE project (VASE, 2021a). The elements will be presented and described in more detail in the following sections:

- The three pillars cover the competency domains that have been identified through desk research reports on values in design and were found to be central when educating responsible designers
- The seven learning objectives guide teachers when teaching for values in design
- The curriculum compass contains 20 specific learning outcomes spanning the three pillars and the SOLO taxonomy and help facilitate students' progression from simple to more complex understandings. The 20 learning outcomes are linked directly to the 28 concrete teaching activities developed throughout the project
- The 28 teaching activities materialise, concretise, and integrate the contents of the pillars, overarching learning objectives and specific learning outcomes into step-by-step activities for teaching values in design
- The 12 assessment activities link back to specific teaching activities to support teachers in checking whether the teaching activities' learning outcomes were achieved by the students

In planning and carrying out teaching for values in design teachers can take the following steps using the VASE framework:

1. Teachers decide on the core area(s) that will be the focus of their teaching for values in design (the pillars) and the main outcome(s) that students should obtain (learning objectives) (Table 1).
2. Teachers then navigate the curriculum compass matrix (Figure 3) and decide on the level of knowledge (SOLO taxonomy) and specific learning outcomes

3. Teachers explore and choose between the possible concrete teaching activities that are connected to the selected learning outcomes in the curriculum compass matrix

4. Teachers prepare their teaching session and decide how they would like to assess whether the learning outcomes are obtained. For this, they can choose to use one of the proposed assessment activities that are connected to each of the individual teaching activities

5. Finally, teachers carry out their teaching for values in design in the classroom

In assessing and checking for students' knowledge, skills, and attitudes in relation to values in design, teachers can take the following steps using the VASE framework:

1. After the teaching activity, teachers can ask students to perform one of the proposed assessment activities connected to the specific teaching activity

2. Students carry out the individual steps of the assessment activity while focusing on addressing the learning outcomes of the teaching activity

3. Teachers and/or students go through the students' products focusing on if, in what way and to what extent the learning outcomes of the teaching activity were obtained

4. Finally, teachers can then return to the curriculum compass and its more general learning outcomes, to assess whether students are in need of deeper or more expansive competencies within a particular area (keeping within the same or similar cells within the matrix), in need of more advanced and complex competencies within the area (moving up the SOLO taxonomy within a row or pillar), or are in need of a general and broader understanding of values in design (moving to different cells or other pillars)

Importantly, the above process of planning, carrying out, and checking for students' learning is often not a linear process. Mostly, teachers will go back and forth between

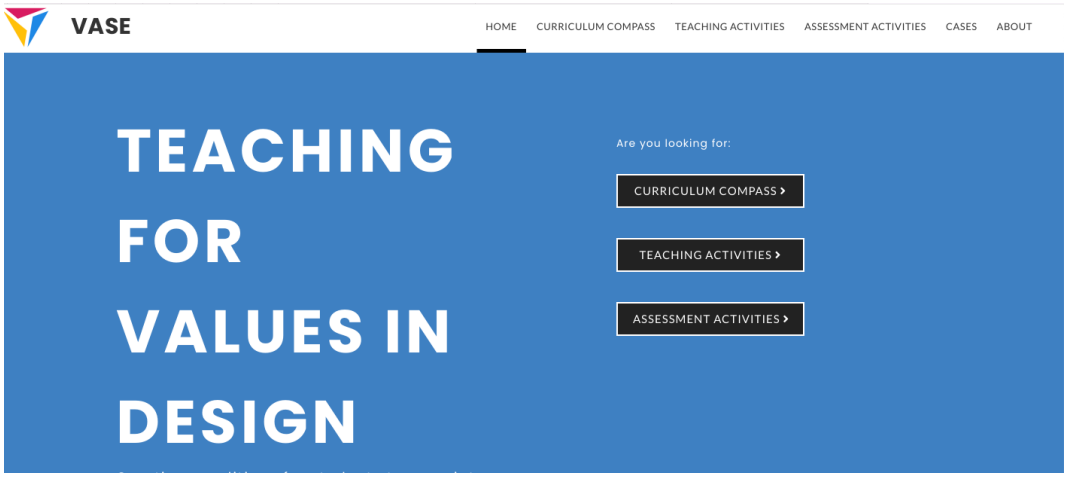


Figure 2. The VASE OER online platform aimed at supporting teachers in teaching for values in design (VASE, 2021).

PILLARS	LEARNING OBJECTIVES
ETHICS AND VALUES	L1. Recognize and describe different values L2. Critically reflect on how values are manifested in designs
DESIGNERS AND STAKEHOLDERS	L3. Identify and describe direct and indirect stakeholders of a design L4. Elicit stakeholder values L5. Identify possible tensions between different stakeholder values and imagine how to mediate these tensions in a design
TECHNOLOGY AND DESIGN	L6. Integrate values into the design process L7. Analyse and critically reflect on the impact of a design (brief) and its manifested values in context

Table 1. Seven key learning objectives of teaching for values in design in relation to the three core competency pillars.

the different steps to explore and reflect on the various opportunities and most relevant learning outcomes or activities to ensure coherence with their way of teaching and the specificities of their discipline, programme, and subject.

As mentioned in the beginning of this section, The VASE framework is comprised of different elements that, when taken together, constitute the main results of the VASE project (VASE, 2021a). These elements will now be presented and described in more detail in the following.

3.1. THE PILLARS

The VASE framework is built up around three core competency pillars:

I) The Ethics and Values pillar explains the underlying theoretical foundations of ethics and values that students need to consider, both in their methods and in their design process, as well as in taking responsibility for their product or service.

II) The Designers and Stakeholders pillar addresses methods and processes for students to ethically engage with different stakeholders and their values, acknowledging that they themselves are stakeholders too.

III) The Technology and Design pillar addresses methods and processes that allow students to practically design and evaluate products and services with values in mind.

The three pillars aim to cover what we consider the main knowledge and skills for becoming a responsible designer: the theoretical background, a focus on different stakeholder needs, as well as the skills to actively engage with technology and values in the design process. Furthermore, the three pillars also cover various parts of the

design process. The Ethics and Values pillar addresses foundational knowledge on different approaches for how to address values in the design process. The Designers and Stakeholders pillar addresses the design phases of research and synthesis. Finally, the Technology and Design pillar covers the knowledge and methods needed for the ideation and evaluation phases.

3.2. THE LEARNING OBJECTIVES

The overarching learning objective guiding the VASE framework is:

to develop the knowledge, skills, and attitudes students need in order to consider the broader context and implications of design and design processes, and through this the possibility for them to become more responsible designers.

This overarching learning objective is broken down into more specific learning objectives. A major challenge was to transform ambiguous terms such as 'become responsible designers' to more specific, observable learning outcomes. The aim was to articulate a set of key learning objectives that promote important cognitive, psycho-motor, and attitudinal attributes, and at the same time provide some useful methods of measuring their achievement (Davies, 2000). To achieve this, we applied the Structure of the Observed Learning Outcome (SOLO) taxonomy (Biggs, 1982).

As a result, we have formulated seven key learning objectives (L1–7), each related to one of the three core competency pillars (Table 1). These learning objectives feature a mixture of low-level, mid-level, and high-level outcomes. We believe that students who have achieved all these learning objectives and consciously apply their abilities in both the design process and deliverables are (on their way to becoming) responsible designers.

3.3. THE CURRICULUM COMPASS

The curriculum compass (Figure 3) provides an overview of the theory, structure, and resources that are the backbone of the VASE framework. It describes general learning outcomes for teaching values in design. These learning outcomes are distributed across the three core competency pillars, design phases, and the SOLO taxonomy levels of competency (Biggs, 1982). Progression in students' learning can be defined as moving up in SOLO levels, from unistructural, to multi-structural, relational, and up to extended abstract level as the highest level. Using the SOLO taxonomy visualises how students' competencies in relation to values in design can develop gradually in multiple directions. By combining the learning outcomes listed in the different cells of the curriculum compass, teachers may support students in becoming

Pillars	Design Phase	SOLO Taxonomy Level			
		Unistructural ●	Multistructural ●●	Relational ●●●	Extended Abstract ●●●●
Ethics & Values	Values Theory	Identify values and name approaches to ethics (e.g., consequentialism) and values in design (e.g., Value-Sensitive Design).	List, describe, and combine different approaches to ethics and values in design.	Analyze, compare, and argue for how values are manifested in design.	Interpret, evaluate, and critically reflect on values and their manifestations in design.
Designers & Stakeholders	Research	Recognize and define the notions around researching designers' and stakeholders' roles and values (e.g., indirect & direct stakeholders).	List, characterize, and report on designers' and stakeholders' roles and values.	Elicit, interpret and contrast designers' and stakeholders' roles and values.	Judge, reason about, and critically reflect on designers' and stakeholders' roles and values.
	Synthesis	Recognize and define the notions around interpreting and combining different designer and stakeholder values into a design direction (e.g., value manifesto).	List, characterize, and report on notions around interpreting and combining different designer and stakeholder values into a design direction.	Interpret, adapt, and plan one's design direction based on the identified designer and stakeholder values.	Reason about, reflect on, and criticize the newly developed design direction based on the identified designer and stakeholder values.
Technology & Design	Ideation	Identify methods for ideating with values (e.g., envisioning).	List and describe methods for ideating with values.	Adapt and apply methods for ideating with values.	Critically reflect on the results of the ideation with values.
	Evaluation	Identify methods for evaluating designs in terms of values (e.g., public evaluation).	List and describe methods for evaluating designs in terms of values.	Adapt and apply methods for evaluating designs in terms of values.	Critically reflect on the evaluation of the designs in terms of values.

Figure 3. The curriculum compass. Source: https://teachingforvaluesindesign.eu/curriculum_compass.html
Continues in the next page.

more responsible designers by developing knowledge, skills, and attitudes that can be deep and specialised (moving students' competencies from unistructural to extended abstract) or wide-ranging and generalist (cutting across all three pillars). Or both. The teacher can choose the direction that fits their purpose for teaching values in design as well as on the aims and objectives of the discipline, programme, module, or teaching session in question.

The 20 learning outcomes described in the curriculum compass are connected to a number of practical teaching activities. The teaching activities are, in turn, connected to suggested assessment activities that support teachers in judging whether the learning outcomes were achieved.

3.4. THE TEACHING ACTIVITIES

The 28 teaching activities included in the VASE framework (Table 2) are connected to specific learning outcomes in the curriculum compass (Figure 3) aimed at creating conditions for students to achieve those learning outcomes. For instance, if a teacher is teaching a course that focuses on concept generation and prototyping (the pillar Technology and Design) and wants their students to be able to 'critically reflect on the results of the ideation with values' (Ideation phase, Extended Abstract level) as a learning outcome, they are offered four different teaching activities: T21. Identifying and Resolving Value Tensions, T22. Exploring Values through Extreme Worlds, T23. Re-designing for Different Cultures, or T24. Envisioning Future Scenarios.

PILLAR	TEACHING ACTIVITIES
I) Ethics and Values	T1. Introduction to values in design T2. Introduction to ethics in design T3. Introduction to cultures and values in design T4. Design with and for certain philosophies T5. Manifestos on values and ethics T6. Values manifested in products, system, and services T7. Values clustering for developing students' value vocabularies T8. Understanding values changing over time
II) Designers and Stakeholders	T9. Individual designer's values identification and hierarchy T10. Design team's values identification and hierarchy T11. Design team's value statement manifesto T12. Listing stakeholders and their values T13. Stakeholder values elicitation T14. Mapping stakeholder value landscapes T15. Project values identification T16. Value-based reformulation of the design draft T18. Constructing value-based design requirements T17. The game changer
III) Technology and Design	T19. Visualising values in design with mood boards T20. Understanding value tensions T21. Identifying and resolving value tensions T22. Exploring values through extreme worlds T23. Re-designing for different cultures T24. Envisioning future scenarios T25. Contextualising values through reflection in action T26. Evaluating values in design with stakeholders T27. Public examination of values in design T28. Design after evaluation of prototype

Table 2. 28 teaching activities distributed across the three core competency pillars.

Each teaching activity contains step-by-step instructions and ready-to-use tools (e.g., lecture slides, worksheets, prompts, readings) so that teachers can easily use it without much preparation. The teaching activities come in various forms including lectures, hands-on design studios, and fieldwork, and covers a wide range of ethical topics including value tensions, value dynamics, non-Western cultures, and extreme worldviews. To ensure consistency across the teaching activities, a pedagogical pattern template was used (Table 3). The activities were quality-assured through multiple rounds

PATTERN TEMPLATE	EXAMPLE OF A TEACHING ACTIVITY
TITLE The title of the specific activity	T24. Envisioning future scenarios
SUMMARY A short summary of the activity focusing on how the activity develops values in design and which competencies the activity is developing	In this teaching activity, students will generate future scenarios to imagine and analyse potential widespread consequences, long-term effects, and societal impacts of their own or others' designs. The activity will lead students to envision at least one use or user scenario that goes beyond what they would normally describe as the intended use of their design. By applying their understanding of potential consequences and effects, they may rethink their designs and design decisions.
BACKGROUND A description of why the values in design competencies are important and what problematic situations during or after their education that students might find themselves in if they do not obtain these competencies	When focusing on users and user experiences, students may approach their own or others' designs from a single, narrow perspective without realising its potential impact on a broader society. Evidently, designs can have widespread consequences and long-term effects on various stakeholders beyond the stakeholders initially imagined, both in positive and negative ways […] (see full text on the VASE OER online platform)
LEARNING OUTCOMES The results and benefits of the activity in the form of specific learning outcomes and results of the activity are described and listed	After the teaching activity students will be able to: ● generate future scenarios to imagine and analyse potential widespread consequences, long-term effects, and societal impacts of their own or others' designs ● apply their understanding of potential widespread consequences and long-term effects to potentially rethink their design and design decisions
PREPARATION Description of how the teacher prepares for the activity and provides the teacher with the materials needed for carrying out the activity	● Prepare a handout of the provided envisioning prompts ● Prepare a short introduction lecture on the importance of being conscious of the broad impact and long-term effects of a design (see e.g., the teaching activity Introduction to Values in design) […] (see full text on the VASE OER online platform)
CONTENT AND STEP-BY-STEP INSTRUCTIONS The activity is described in step-by-step instructions for the teacher to support the teacher in carrying out the activity and facilitate students' learning so they might achieve the activity's learning outcomes	1. Give the lecture that you have prepared. Ask the students to select a project that they are developing as a group or that they have developed on their own or present an example design case that they can develop a scenario for 2. Walk through the process of the activity, instructions, time plan, and envisioning prompts. Describe the expected outcome, which is at least one future scenario for the design, using one or more of the envisioning criteria, and a reflection on possible consequences for the design […] (see full text on the VASE OER platform)
SUGGESTED ASSESSMENT ACTIVITIES AND CRITERIA Two suggested assessment activities are described and a link to the assessment activities are given to support the teacher in choosing relevant assessment activities when checking whether the intended learning outcomes are achieved by the students Also, bullet points describing what students might focus on when performing the assessment activity are given to support both teachers and students in obtaining the learning outcomes of both teaching activity and curriculum compass	To assess whether the intended learning outcomes were attained by the teaching activity the following assessment activities can be carried out (in class or after class). Suggested assessment activities: Assessing students' learning by asking them to apply ***their learning about future scenarios on a case study*** (summative assessment) by imagining and analysing potential consequences, long-term effects, or societal impacts of a design through a value scenario using relevant envisioning criteria (including values) and prompts. […] (see full text on the VASE OER platform) Suggested assessment criteria: In the assessment activity ask students to focus on ● describing what envisioning criteria and prompts are relevant to apply for a specific case ● imagining potential consequences, long-term effects, and societal impacts of a design through a value scenario that goes beyond what would normally be described as intended use, using relevant envisioning criteria (including values) and prompts ● analysing the potential consequences of a design using relevant envisioning criteria (including values) through a value scenario and providing suggestions for how to mitigate negative consequences (e.g., in regard to re-design, further stakeholder dialogue, possible tensions) through rethinking the design.

Table 3. The pattern template, here filled out with parts of the teaching activity T24. Envisioning future scenarios.

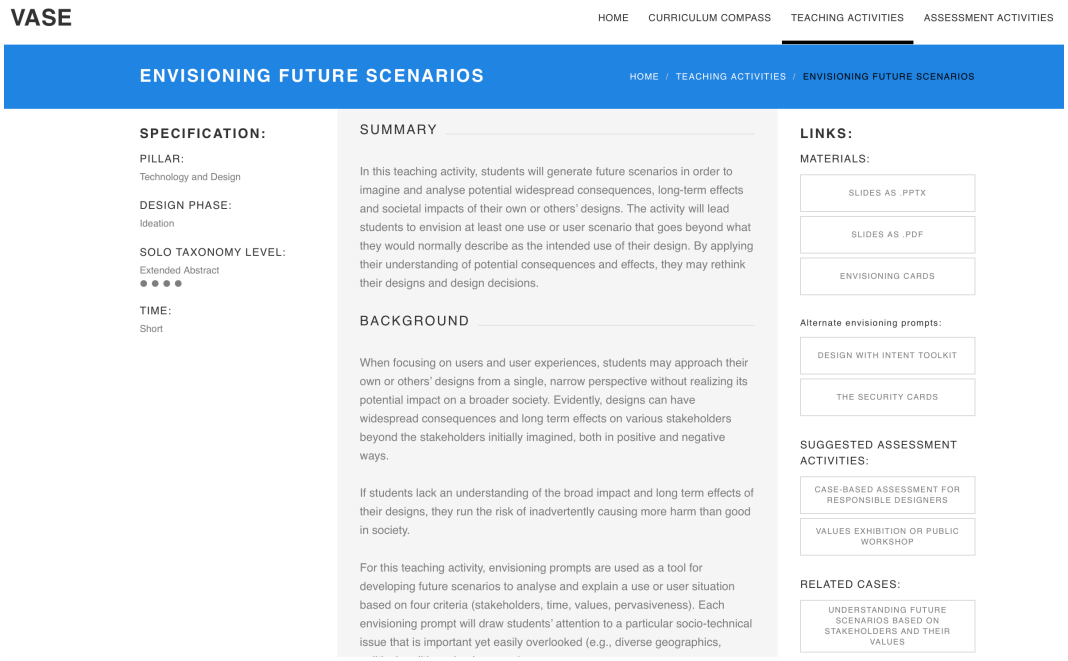


Figure 4. Screenshot from the VASE OER online platform presenting the teaching activity T24. Envisioning future scenarios (VASE, 2021). Source: https://teachingforvaluesindesign.eu/24_envisioningfuturescenarios.html

of peer-review using the pedagogical pattern method of shepherding (Harrison, 1999).

To help teachers to navigate the full collection of teaching activities published on the VASE OER online platform (VASE, 2021), the activities can be filtered based on their core competency pillar, design phase, time needed to perform the activity and SOLO level (Figure 4).

3.5. THE ASSESSMENT ACTIVITIES

In educating responsible designers, we do not find it sufficient to consider only the final material result of learning (e.g., the design artefact). Rather, the learning process needs to be considered in assessment. This is especially important for teaching values in design, where the final deliverables (e.g., the way a prototype looks and functions) are arguably less important than the process used to arrive at this material result (e.g., properly considering stakeholders and their values in the design process). The teaching activities included in the VASE framework are therefore accompanied by assessment activities with a focus on learning processes. The framework includes 12 assessment activities (Table 4), which are structured around the three main competence types related to learning, namely: knowledge, skills, and attitudes (Baartman & Bruin, 2011). Additionally, the assessment activities are structured in four assessment forms: summative, formative, authentic, and ipsative.

In summative assessment, the focus is on measuring the level of learning, typically against standardised criteria, by collecting, interpreting, and reporting evidence

of learning (Dolin, 2017). In formative assessment, the goal is to provide feedback that moves students forward (Hughes, 2014). Theoretically, any learning outcome can be assessed summatively (typically by giving a grade) or formatively (typically by providing feedback). This includes traditional assessment activities such as examinations or (prototype) design deliverables.

Authentic assessment (or performance assessment), is an assessment form which focuses on the value of students' learning in the 'real world' (situated assessment in context), translating school-based ideas to authentic situations and tasks (Ashford-Rowe et al., 2014). Ipsative assessment activities aim to activate students as owners of learning (Hughes, 2014). Ipsative assessment compares a learners' current performances with their previous performances, making it a highly personalised form of assessment. Throughout this process, ipsative assessments provide valuable feedback for students about their strengths and weaknesses.

Each of the 28 teaching activities included in the VASE framework are linked to two assessment activities. For example, the teaching activity T24. Envisioning future scenarios is linked to the two assessment activities A9. Case-based assessment for responsible designers and A12. Value-based public exhibition or public workshop (Figure 5).

The assessment activities were developed using the same iterative method, peer-review process, and similar pattern structure as for the teaching activities. The more

COMPETENCY TYPE	ASSESSMENT FORM	ASSESSMENT ACTIVITIES
Knowledge	Summative	A1. Mind mapping for responsible design
	Formative	A2. Reflective values report
	Ipsative	A3. Personal video
	Authentic	A4. Applying knowledge to real-world examples
Skills	Summative	A5. Video pitching for responsible designers
	Formative	A6. Round Robin values Brainwriting
	Ipsative	A7. Historical value timeline
	Authentic	A8. Peer feedback for responsible designers
Attitude	Summative	A9. Case-based assessment for responsible designers
	Formative	A10. Self-assessment for responsible designers
	Ipsative	A11. Blogging for responsible designers
	Authentic	A12. Values exhibition or public workshop

Table 4. 12 assessment activities spanning over competency types and assessment forms.

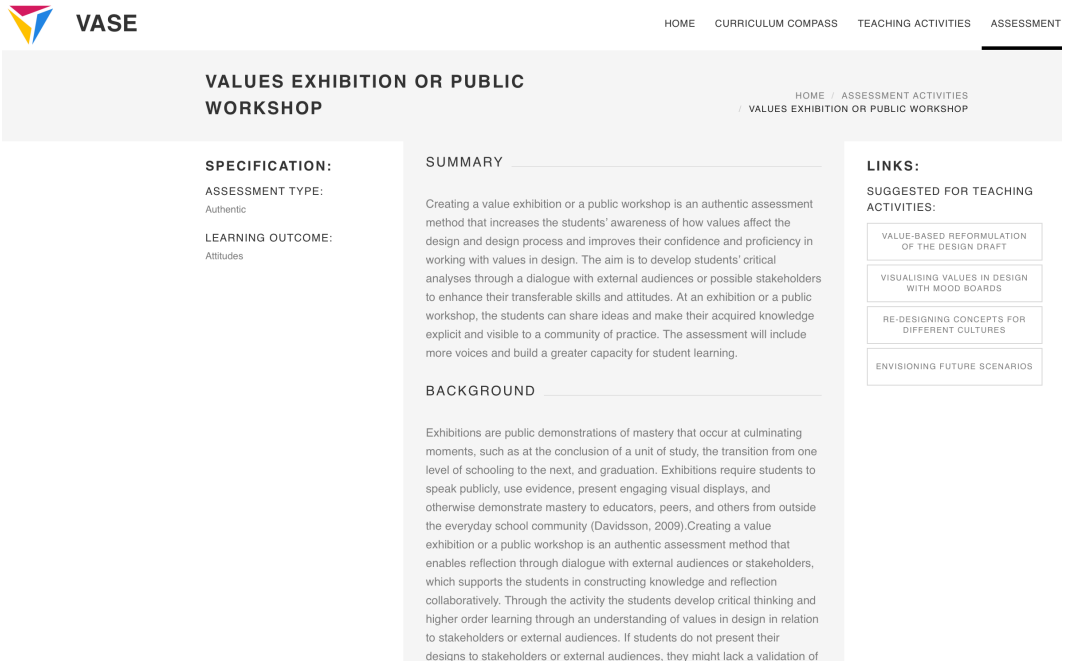


Figure 5. Screenshot from the VASE OER online platform presenting the assessment activity A12. Values exhibition or public workshop (VASE, 2021).

general description of the assessment activity was then developed into more specific assessment criteria formulated in each teaching activity (see the last row in Table 3 for examples on 'Suggested assessment criteria').

3.6. USING THE VASE PEDAGOGICAL FRAMEWORK

The VASE framework can be used both as a reflective planning tool for targeting specific value-sensitive competencies or learning outcomes, as well as an inspirational resource for insight into how one can teach for values in design that can be adopted across different educational contexts. In this way, the VASE framework can, on the one hand, support teachers in systematically planning and carrying out teaching, and, on the other hand, help facilitate robust learning by giving teachers ways to assess and check for learning.

The VASE framework can be used to:

- Select specific stand-alone activities

Teachers can explore the overarching learning objectives and select those that are most relevant to their discipline, curriculum, or course. The learning objectives are described in broad terms while the teaching activities connected to each of the learning objectives execute them in concrete ways

- Create in-depth learning pathways

Teachers can combine concrete teaching activities that move students from a

simple (unistructural) to a complex (extended abstract) level of understanding of values in design within a specific pillar. Each of the three pillars represent core areas of teaching values in design. Integrating learning pathways in teaching allows students to build deeper knowledge, skills, and attitudes progressively to acquire a desired set of competencies within a specific pillar

- Give students a broad foundation

Teachers can combine concrete activities across all three pillars that create a broad foundation for students to become more responsible designers. Creating a broad foundation within a certain level of understanding allows students to develop a more holistic approach to values in design in relation to a select level of competence

4. CONCLUDING WORDS

This chapter puts forward a proposal for a pedagogical framework, namely the VASE framework (Figure 1), for educating responsible designers by teaching values in design. The VASE framework is unique in a European higher education context, both in its systematic use of the adapted pedagogical pattern method and double-pedagogical framework to create research-based, practice-informed, and peer-reviewed teaching activities, as well as its open educational platform for teaching and assessment activities. We will end this chapter by reflecting upon gaps and opportunities, which open up for further explorations and development of educational resources creating conditions for students to grow into responsible designers.

4.1. THE DEVELOPMENT OF THE VASE FRAMEWORK

In the iterative development of the VASE framework, we have strived to include and cover as many pedagogical aspects as possible. However, as made visible in the model of the VASE framework (Figure 1), there are some gaps. For instance, there are no teaching or assessment activities connected to the lowest unistructural level of the learning outcomes in the curriculum compass. This is quite deliberate, as it has never been a goal of this work to support the lowest levels of the SOLO taxonomy. The approach has been to activate the students' learning in practice, meaning that the students should be able to not only identify various relevant aspects, but also be able to describe why they are relevant. A further gap in the framework is in the third pillar (Technology and Design), where there are no activities in either the unistructural or the multi-structural level. The reason for this is that in these later phases of the design process, the learning progression should have moved up to relational and extended

levels, where the student is able to adapt, apply, and reflect on various aspects for values in design. In line with this, it is also evident that in the first pillar (Ethics and Values) there is only one single teaching activity in the extended abstract level, namely T8. Understanding values changing over time. As this is in the first pillar, covering the underlying knowledge in the beginning of the design process, this is only expected that most activities are in the multi-structural and relational levels.

The SOLO taxonomy implies that learning can be divided into four categories, which arguably is a reductionist view on learning (Wong, 2007; Khan, 2015). We witnessed this in the development process, where it was sometimes difficult to fit multi-layered learning outcomes into a single SOLO category. There are, however, many good arguments for using the SOLO taxonomy; it is already widely used in education and is a familiar tool for teachers (Wong, 2007). SOLO provides an established and a common language for articulating and discussing learning outcomes (Hook, 2016), and that has been applied to many different subjects, from poetry and history (Biggs & Collis, 1982) to mathematics (Hattie & Brown, 2004). Further, its simplicity allows learning outcomes to be formulated precisely and in ways that make them comparable between educational contexts.

Taken together, the strong research foundation in the form of the desk research reports, the international co-development of activities, as well as the rigorous, consistent, and shared method and practises enabled the VASE project to achieve its ambition to deliver a high-quality open educational resource for the development of student' competencies for values in design and for promoting the role of teaching values in design. However, as it is recently launched and, so far, only partly tested in four countries, further investigation and more large-scale piloting are needed to assess whether the VASE framework is applicable across broader educational and cultural contexts.

4.2. APPLYING THE VASE FRAMEWORK: HOW DOES IT WORK IN PRACTICE?

In designing open educational resources, a high level of generalisation is often necessary to accommodate a wide range of learning contexts, as well as to support a high degree of appropriation. Due to the situated nature of learning and teaching processes, teaching activities might not be used in the exact same way that the educational designers had originally envisioned. This appropriation is not a sign of failure, but rather shows that the users (that is, teachers) understand and are comfortable enough with the design to try using it in their own ways (Dix, 2007).

We have conducted a comparative study, involving two teachers in two different countries and educational settings, where they piloted the teaching activity T7. Values clustering for developing students' value vocabularies (Nilsson et al., 2020).

Here, we have gained an understanding of the importance of including several inspirational examples for the teachers to see how one can perform a teaching activity in a flexible manner, instead of prescribing a strict guideline for implementing the activity. What became evident in our study is how teachers can take the same teaching activity description and then be creative to modify it to meet their own needs. That is why a design for use must be about a design for change (Dix, 2007), and why we have applied an adaptability approach when developing the activities. Adaptability here refers to describing the activities so that they contain enough information and instruction for non-experts to carry them out but are open and flexible enough to be adapted across different disciplines or contexts and for different pedagogical purposes or approaches. The possibility of appropriation can further create a feeling of ownership since the users will feel that they are in control and can experiment to find new ways to achieve their goals. However, since the VASE framework and OER have been launched only recently, studying the long-term appropriation would be an interesting topic for future studies, tracing the iterative modification of a design activity initiated by the same teacher over time.

4.3. EDUCATING THE RESPONSIBLE DESIGNER

As part of the iterative development of the teaching materials, we have involved teachers and students in piloting the material. The teachers reported that they experience professional development through a qualitative and needed update of the design curriculum, and increased capacity to teach for values in design in relevant and innovative ways. The participating students have reported that they experience achieving relevant labour market skills, future-ready education, and updated knowledge about how to understand and act ethically on trade-offs between values, design, and social forces that emerge through human use of that design. So, in that sense, early results indicate that we have achieved what we set out to do: educate responsible designers.

In a recent study we performed an adapted version of the teaching activity T24. Envisioning future scenarios, to identify the large-scale effects of teaching values in design (Kok et al., 2021). We first developed a traditional scenario-based design (SBD) to assess the implications of teaching values in design (Rosson & Carroll, 2002). This is not typically part of the envisioning teaching activity but allowed us to make a comparison between the SBD approach and the value scenario approach. We then developed two value scenarios, as described by Nathan et al. (2008), using prompts divided into the four envisioning topics to guide us (Friedman & Hendry, 2012). While the traditional scenario considers mostly the immediately obvious and desirable consequences of teaching values in design for direct stakeholders, the value scenarios – by incorporating direct and indirect stakeholders, time, values, and pervasiveness – open our eyes to

less obvious, unintended, concrete, long-term, and large-scale effects, both good and bad. It demonstrates that individual classroom outcomes are not the only important consequences one's teaching may have. Rather, the way education shapes students continues to play out beyond the classroom and throughout their professional lives. Envisioning has helped clarify in what ways students as well as indirect stakeholders (such as the people for whom they create designs) could be affected by teaching. At the same time, as designers of educational resources, we are aware of the fact that over time the political significance of education will change, and that we can never envision and imagine the full implications of our project (Tromp et al., 2011; Winner, 1980; Kok et al., 2021). While we see the VASE framework as a first but important step in offering a reflective pedagogical framework on teaching values in design, there is still much work ahead. This includes trying out the teaching material in other cultural and educational contexts, by other disciplines, and even more importantly, to try it out in contexts with other value systems, beyond the European and Western world. Hopefully the VASE framework and the teaching resources can continue to grow and travel widely over time. At least we, as authors, do not see this work as completed, and we invite the community to contribute and to help us to keep it relevant.

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